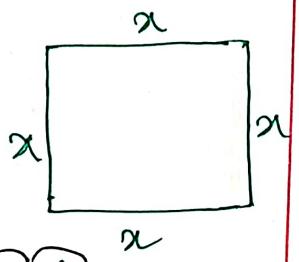
-: 12th BOARD HACKER JUGAADU NOTES :-

Application of Derivative

1) Derivative as a rate of change
2) Monotonicity
3) Maxima & Minima

Derivative as a Rate of change

$\frac{dy}{dx} \rightarrow$

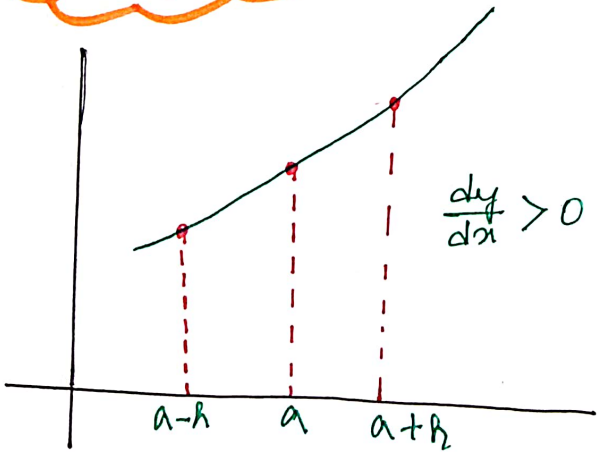


$A = x^2$

diff wrt t

$\frac{dA}{dx} = 2x \quad \frac{dA}{dt} = 2x \cdot \frac{dx}{dt}$

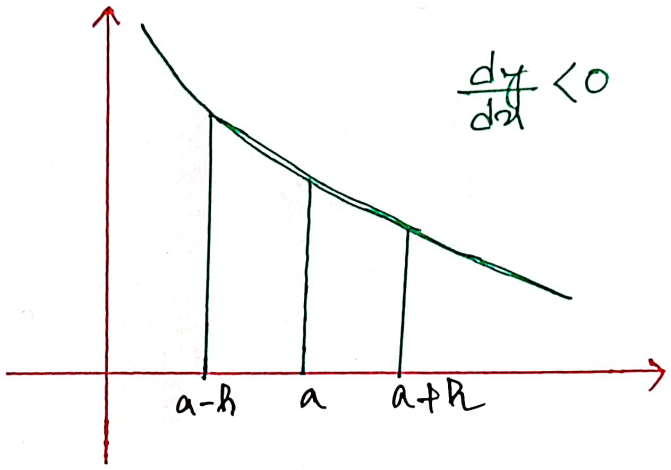
Monotonicity at a point $x=a$



A function $f(x)$ is called an increasing function or monotonically increasing function at point $x=a$, if -

$f(a-h) < f(a) < f(a+h)$ where

$h \rightarrow 0^+$



A function $f(x)$ is called an decreasing function or monotonically decreasing function at point $x=a$, if -

$f(a-h) > f(a) > f(a+h)$ where

$h \rightarrow 0^+$

Test for Monotonicity

Test for monotonicity in a differentiable function at a point.

ii) If $f'(x)$ changes sign from positive to negative while passing through $x=a$ from left to right then $x=a$ is a point of local maxima.

iii) If $f'(x)$ does not changes its sign about $x=a$ then $x=a$ is neither a point of maxima nor minima.

Note :-

1. This test is applicable only for continuous functions.

Second Derivative Test

If $f(x)$ is a differentiable function in the neighbourhood of $x=a$ such that $f'(a)=0$ and $f''(a)$ exists then we can predict maxima or minima at $x=a$ by examining the sign of $f''(a)$

i) If $f''(a)>0$ then $x=a$ is a point of local minima

ii) If $f''(a)<$ then $x=a$ is a point of local maxima.

iii) If $f''(a)=0$ then Second derivative test does not give us conclusive results.

i) If $f'(a) > 0$ then $f(x)$ is increasing at $x = a$

ii) If $f'(a) < 0$ then $f(x)$ is decreasing at $x = a$.

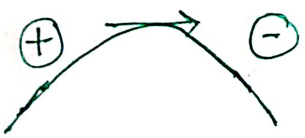
iii) If $f'(a) = 0$ then examine the sign of $f'(a^+)$ and $f'(a^-)$.

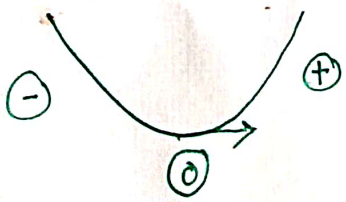
a) If $f'(a^+) > 0$ and $f'(a^-) > 0$ then $f(x)$ is monotonically increasing at $x = a$.

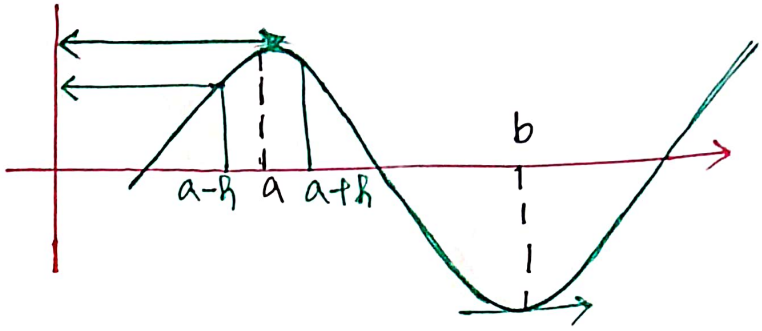
b) If $f'(a^+) < 0$ and $f'(a^-) < 0$ then $f(x)$ is monotonically decreasing at

c) Otherwise neither increasing nor decreasing at $x = a$ i.e. non-monotonic at $x = a$.

Maxima - Minima







$x = a$ is pt of local maxima
$x = b$ is pt of local minima.

for maxima

$f(a) > f(a+h)$

& $f(a) > f(a-h)$

$f'(x) = 0$ at local maxima & minima

$f'(a) = 0$ & $f'(b) = 0$

First Derivative Test





i) If $f'(x)$ changes sign from negative to positive while passing through $x = a$ from left to right then $x = a$ is a point of local minima.